

## Chapter 2 Assignment Due date: March 30, 2026 (Mon)

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### 2.1

Sol.

i.

$$f(x) = F'(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 1, & \text{o.w.} \end{cases}$$

ii.

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f(x) dx = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x dx \\ &= \frac{1}{b-a} \left[ \frac{x^2}{2} \right]_a^b = \frac{1}{b-a} \cdot \frac{b^2 - a^2}{2} = \frac{a+b}{2} \end{aligned}$$

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x^2 dx \\ &= \frac{1}{b-a} \left[ \frac{x^3}{3} \right]_a^b = \frac{1}{b-a} \cdot \frac{b^3 - a^3}{3} = \frac{a^2 + ab + b^2}{3} \end{aligned}$$

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{a^2 + ab + b^2}{3} - \left( \frac{a+b}{2} \right)^2 = \frac{(b-a)^2}{12}$$

- iii.
- The pdf is constant on  $[a, b]$ , so all values in the interval are equally likely.
  - The graph of pdf is a rectangle over the interval  $[a, b]$ .

iv.

$$P(7 \leq X \leq 9) = \frac{9-7}{10-5} = \frac{2}{5} = 0.4$$

### 2.6

Sol.

$$X \sim \text{Bin}(n=10, p=0.2)$$

$$f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

i.

$$E(X) = np = 10 \cdot 0.2 = 2$$

$$Var(X) = np(1 - p) = 2 \cdot (1 - 0.2) = 1.6$$

ii.

$$P(3 \leq X \leq 5) = P(X = 3) + P(X = 4) + P(X = 5)$$

$$= \binom{10}{3}(0.2)^3(0.8)^7 + \binom{10}{4}(0.2)^4(0.8)^6 + \binom{10}{5}(0.2)^5(0.8)^5$$

$$\approx 0.3158$$

## 2.8

**Sol.**

10, 12, 15, 17, 20, 20, 23, 25, 28, 30, 40, 43, 45, 48, 50, 50, 59, 61, 64, 70.

i.  $20 \cdot \frac{1}{4} = 5 \Rightarrow Q_1 = \frac{a_5 + a_6}{2} = \frac{20 + 20}{2} = 20$

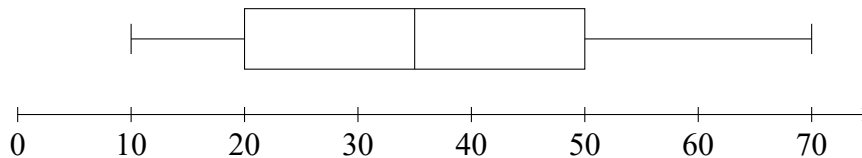
$20 \cdot \frac{1}{2} = 10 \Rightarrow Me = \frac{a_{10} + a_{11}}{2} = \frac{30 + 40}{2} = 35$

$20 \cdot \frac{3}{4} = 15 \Rightarrow Q_3 = \frac{a_{15} + a_{16}}{2} = \frac{50 + 50}{2} = 50$

ii.  $IQR = Q_3 - Q_1 = 50 - 20 = 30$

iii. Lower bound =  $Q_1 - 1.5IQR = 20 - 45 = -25$

Upper bound =  $Q_3 + 1.5IQR = 50 + 45 = 95$



## 2.9

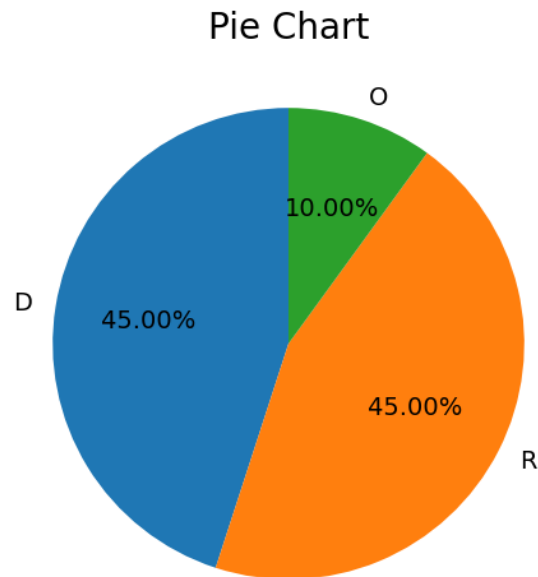
**Sol.**

D, D, R, O, R, R, R, D, R, D, D, R, D, D, R, R, R, O, D, D

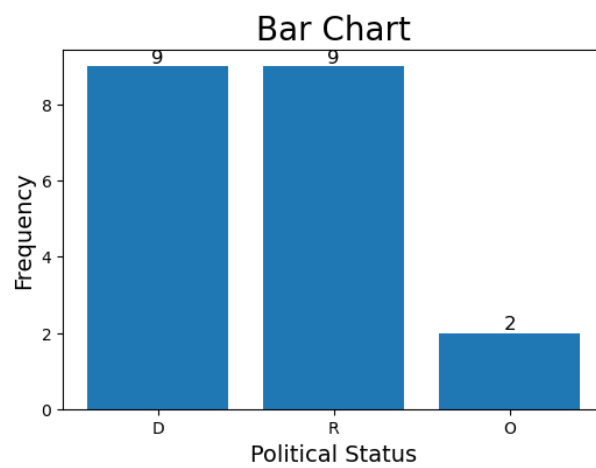
i.

| Political Status       | Frequency | Relative Frequency |
|------------------------|-----------|--------------------|
| <i>D</i> (Democrats)   | 9         | 0.45               |
| <i>R</i> (Republicans) | 9         | 0.45               |
| <i>O</i> (Others)      | 2         | 0.10               |

ii.



iii.



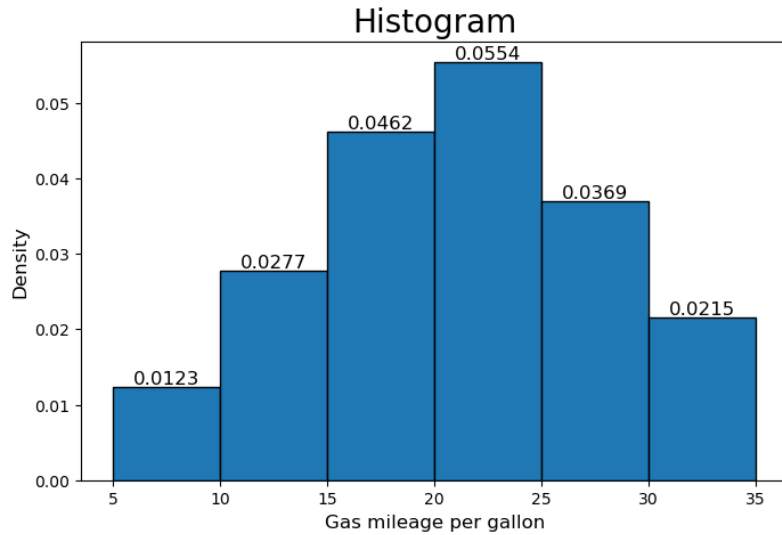
## 2.11

**Sol.**

i.

| Class Interval | Frequency | Relative Frequency | Density |
|----------------|-----------|--------------------|---------|
| [5, 10)        | 20        | 0.0615             | 0.0123  |
| [10, 15)       | 45        | 0.1385             | 0.0277  |
| [15, 20)       | 75        | 0.2308             | 0.0462  |
| [20, 25)       | 90        | 0.2769             | 0.0554  |
| [25, 30)       | 60        | 0.1846             | 0.0369  |
| [30, 35)       | 35        | 0.1077             | 0.0215  |

ii.



iii.

$$P(5 \leq X < 20) = \frac{20 + 45 + 75}{325} \approx 0.4308$$

## 2.14

**Sol.**

i.

$$\hat{\pi} = \frac{x}{n} = \frac{272}{400} = 0.68$$

ii. 95% CI for  $\pi$  is

$$\hat{\pi} \pm Z_{0.025} \sqrt{\frac{\hat{\pi}(1 - \hat{\pi})}{n}} = 0.68 \pm 1.96 \sqrt{\frac{0.68 \cdot 0.32}{400}} = 0.68 \pm 0.0457$$

So the 95% CI is

$$(0.6343, 0.7257)$$

## 2.19

**Sol.**

| Year              | 1942 | 1948 | 1954 | 1959 | 1967 | 1970 | 1975 | 1981 | 1983 |
|-------------------|------|------|------|------|------|------|------|------|------|
| City A            | 45   | 0    | 4    | 21   | 9    | 1    | 30   | 17   | 53   |
| City B            | 40   | 10   | 2    | 20   | 7    | 9    | 33   | 17   | 50   |
| $d_i = A_i - B_i$ | 5    | -10  | 2    | 1    | 2    | -8   | -3   | 0    | 3    |

i. We test

$$H_0 : \mu_d = 0 \text{ vs } H_1 : \mu_d \neq 0$$

The sample mean of the differences is

$$\bar{d} = \frac{5 - 10 + 2 + 1 + 2 - 8 - 3 + 0 + 3}{9} = -\frac{8}{9} \approx -0.8889$$

and the sample standard deviation is

$$s_d = \sqrt{\frac{\sum(d_i^2) - (\sum d_i)^2/n}{n-1}} = \sqrt{\frac{(5^2 + \dots + 3^2) - [(-8)^2/9]}{8}} \approx 5.1099$$

Thus, the test statistic is

$$t = \frac{\bar{d} - \mu_d}{s_d/\sqrt{n}} = \frac{-0.8889}{5.1099/\sqrt{9}} \approx -0.5219$$

The degree of freedom are  $9 - 1 = 8$  For a two-sided test with  $\alpha = 0.05$ , the critical value  $t_{0.025, 8} = 2.306$  Since

$$-0.5219 < 2.306$$

So we fail to reject  $H_0$ .

Therefore, at the 0.05 significance level, there is no significance differences between the average snowfalls of the two cities.

## 2.21

**Sol.**

5.6, 2.5, 3.7, 7.9, 11.3, 6.4, 3.2, 2.7, 0.5, 5.7, 3.2, 2.4, 7.6, 3.8, 5.7, 5.8, 3.4, 5.1, 2.2, 4.0, 4.2, 4.2, 5.5, 1.5, 1.6, 6.3, 4.3, 5.0, 8.6, 6.3, 4.3, 6.4, 4.6, 2.0, 4.4

i. Let

$$g(\mu) = \mu(\mu + 1) = \theta$$

$$n = 35, \bar{X} = \frac{5.6 + \dots + 4.4}{35} \approx 4.6257, s^2 = \frac{(5.6^2 + \dots + 4.4^2) - [4.6257^2/35]}{35 - 1} \approx 2.2203$$

The point estimate of  $\theta$  is

$$\hat{\theta} = g(\bar{X}) = \bar{X}(\bar{X} + 1) = 4.6257(4.6257 + 1) \approx 26.0229$$

Since

$$g'(\mu) = 2\mu + 1$$

By delta method, we have

$$\sqrt{n} (g(\bar{X}) - g(\mu)) \rightarrow N(0, [g'(\mu)]^2 \sigma^2)$$

So the approximate standard error is

$$\widehat{SE}(g(\bar{X})) = \sqrt{\frac{(2\bar{X} + 1)^2 s^2}{n}} = \sqrt{\frac{(2 \cdot 4.6257 + 1)^2 2.2203^2}{35}} \approx 3.8473$$

95% CI for  $\theta$  is

$$\hat{\theta} \pm 1.96 \widehat{SE} = 26.0229 \pm 1.96 \cdot 3.8473 = 26.0229 \pm 7.5407$$

So the 95% CI is

$$(18.4822, 33.5637)$$

```

ii. x = np.array([
    5.6, 2.5, 3.7, 7.9, 11.3, 6.4, 3.2, 2.7, 0.5, 5.7,
    3.2, 2.4, 7.6, 3.8, 5.7, 5.8, 3.4, 5.1, 2.2, 4.0,
    4.2, 4.2, 5.5, 1.5, 1.6, 6.3, 4.3, 5.0, 8.6, 6.3,
    4.3, 6.4, 4.6, 2.0, 4.4])

B = 1000
n = len(x)
np.random.seed(611211106)

def theta_from_mean(m):
    return m * (m + 1)

theta = np.empty(B)

for b in range(B):
    sample = np.random.choice(x, size=n, replace=True)
    mean = np.mean(sample)
    theta[b] = theta_from_mean(mean)

lower = np.percentile(theta, 2.5)
upper = np.percentile(theta, 97.5)

print(f"95% bootstrap CI for theta = ({lower:.4f}, {upper:.4f})")
---
95% bootstrap CI for theta = (19.5137, 34.2572)

```

Using the bootstrap method, the 95% bootstrap CI is

(19.5137, 34.2572)

The delta-method interval is close to the bootstrap interval, and it is slightly shifted to the right.

## 2.25

**Sol.**

The kernel density estimator of  $f(x)$  is defined by

$$\hat{f}(x) = \frac{1}{n} h_n \sum_{i=1}^n K\left(\frac{x - X_i}{h_n}\right)$$

where  $h_n > 0$  is a bandwidth and  $K$  is a kernel function. Therefore

```

data = np.array([
    5.6, 2.5, 3.7, 7.9, 11.3, 6.4, 3.2, 2.7, 0.5, 5.7,
    3.2, 2.4, 7.6, 3.8, 5.7, 5.8, 3.4, 5.1, 2.2, 4.0,
    4.2, 4.2, 5.5, 1.5, 1.6, 6.3, 4.3, 5.0, 8.6, 6.3,
    4.3, 6.4, 4.6, 2.0, 4.4])
n = len(data)
h = 1.0

```

```

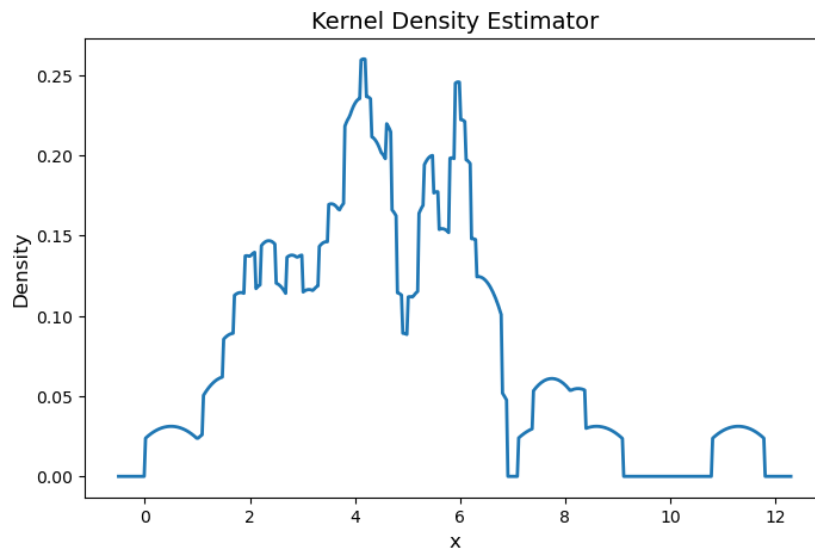
# Epanechnikov kernel
def epanechnikov(u):
    return np.where(np.abs(u)<=0.5, 12/11*(1-u**2), 0.0)

def kde_epanechnikov(x, data, h):
    u = (x[:, None]-data[None, :])/h
    return np.sum(epanechnikov(u), axis=1)/(len(data)*h)

x_grid = np.linspace(min(data)-1, max(data)+1, 500)
y_kde = kde_epanechnikov(x_grid, data, h)

plt.figure(figsize=(8, 5))
plt.plot(x_grid, y_kde, linewidth=2)
plt.xlabel('x', fontsize=12)
plt.ylabel('Density', fontsize=12)
plt.title('Kernel Density Estimator', fontsize=14)
plt.show()

```



**2.26**

**Sol.**